

THIAGO LAIDLER VIDAL CUNHA

Quantitative Analyst — Model Validation, ML Infrastructure & Data Risk

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SUMMARY

Quantitative analyst with a data/ML engineering background transitioning into risk: 2.5+ years building and controlling production model and data infrastructure at a CVM-regulated asset manager, plus M.Sc. research on ML classification of rare events in noisy time series ($F1 \approx 0.97$ on held-out data). Hands-on with cloud ML pipelines (Azure), data quality and auditability controls, model monitoring, and out-of-sample validation with strict leakage controls, including a documented case of a validation pipeline correctly rejecting its own strategy. Experience building LLM-based tools and AI-assisted engineering workflows. Fluent English; co-author of a peer-reviewed paper (MNRAS, Oxford Academic).

TECHNICAL SKILLS

- **Programming:** Python (pandas, NumPy, SciPy, scikit-learn, statsmodels, PyTorch), SQL, Git, Bash
- **ML & data infrastructure:** cloud pipelines (Azure Functions, Cosmos DB, SQL Server), ETL design, data quality and auditability controls, reproducible environments and versioned configs
- **Model validation & monitoring:** out-of-sample validation (purged/combinatorial CV, deflated Sharpe), leakage and overfitting detection, feature engineering and selection, regime/distribution-shift detection, performance degradation tracking, benchmark null models
- **Risk & governance:** risk and performance analytics, NAV controls and reconciliation with administrators/custodians, compliance reporting, documentation/testing/code-review discipline (pytest, ruff, mypy)
- **AI/LLM:** LLM-based automation and research tools; AI-assisted engineering (Claude Code, Cursor, spec-driven workflows)

EXPERIENCE

Transfero Asset Management — digital-asset manager (CVM-regulated), Rio de Janeiro Jan 2024 – Present

Quantitative Analyst — Data Science & Analytics (Apr 2025 – Present)

- Engineered cloud data pipelines (Azure Functions, Cosmos DB, SQL, Python) powering portfolio and NAV analytics, cutting processing time by 60%+ while strengthening data quality, auditability, controls, and documented data flows (lineage).
- Built automated monitoring dashboards providing continuous visibility into data pipeline health and key performance/risk indicators, used daily by trading, operations, and compliance teams.
- Validated statistical models with robust out-of-sample testing (time-series modeling, cointegration), applying strict controls against look-ahead bias and data leakage.
- Produced risk and performance analytics for a systematic fund turning over ~USD 1B/year, supporting allocation decisions and compliance reporting.

Middle Office Analyst — graduate internship (Jan 2024 – Apr 2025)

- Automated cash-reconciliation and NAV validation controls with fund administrators and custodians (Python, SQL, VBA), reducing closing time and operational risk.
- Designed, built, and operated a data product tracking DeFi liquidity-pool positions across multiple wallets, with automated daily alerts and fund-style quota (NAV) accounting; owned scoping, delivery, and the documented cost-benefit decision to retire it when LLM tooling made custom maintenance uneconomical.

Observatório Nacional (MCTI) — M.Sc. Researcher Mar 2024 – 2026

- Built an end-to-end ML pipeline for rare-event detection in noisy time series with full reproducibility discipline: versioned environments and configs, traceable pipeline audit documentation, feature engineering with 400+ documented features, physics-based synthetic data augmentation, and threshold tuning for high-sensitivity regimes.
- Results: $F1$ 0.978–0.994 and $AUC-ROC > 0.998$ on held-out data; validated on a real out-of-sample event (Quaoar stellar occultation).

Federal University of Rio de Janeiro (UFRJ) — Undergraduate Researcher (PIBIC/CNPq) Oct 2020 – Jan 2023

- Co-author of a peer-reviewed publication in *Monthly Notices of the Royal Astronomical Society* (Oxford Academic).

SELECTED PROJECTS (GITHUB.COM/TLAIDLER)

- **Model validation case study — regime detection in BTC/USDT:** full López de Prado methodology (dollar bars, triple-barrier labeling, purged/combinatorial cross-validation, deflated Sharpe ratio) with strict leakage controls; the pipeline correctly rejected its own strategy out-of-sample and the null result was documented and reported — an exercise in challenging one's own models. Detected market regime shifts (distribution drift) across BTC, S&P 500, and Ibovespa.
- **Typed, tested financial web app:** Flask + Pydantic, pytest suite, linted and type-checked (ruff, mypy), documented architecture and formulas validated against finance literature — engineering-control discipline applied to a personal project.

EDUCATION

- **M.Sc., Astronomy/Astrophysics** (machine learning for time series) — Observatório Nacional (MCTI), 2026
- **B.Sc., Astronomy/Astrophysics** — Federal University of Rio de Janeiro (UFRJ), 2023

LANGUAGES & CERTIFICATIONS

- Portuguese (native); English (fluent, C1 — 1,800h Oxford University Press program)
- CEA — Certified Investment Specialist (ANBIMA);
- Data Science Bootcamp (Santander, 89h)